

PROJECT PROPOSAL

GROUP PROJECT

IIT372-2

Group No: IIT 10

LifeMate

A Daily Life Routine Management System
(Specializing Work From Home Users)

Industrial Information Technology Degree Program

Department of Computer Science and Informatics

Faculty of Applied Sciences

Uva Wellassa University of Sri Lanka

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DECLARATION

We hereby declare that the project will be developed by us and will be our own effort and that no part will be plagiarized without citations under the supervision of Ms.K.A.A.Chathurangi, Lecturer, Department of Computer Science and Informatics, Faculty of Applied Science, Uva Wellassa University of Sri Lanka. This Project Proposal is submitted for the partial fulfillment of the requirement of the course unit IIT372-2, Project II for the degree of Industrial Information Technology.

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1. Introduction

1.1 Project Title

LifeMate: A Daily Life Routine Management System (Specializing Work From Home Users)

1.2 Project Description

Modern lifestyles, especially among work-from-home users, often lead to unbalanced routines, unhealthy habits, and increased stress levels. Individuals working remotely frequently struggle to manage their daily schedules effectively due to the lack of structured environments, making it difficult to balance work, rest, and personal health.

This project proposes a system that tracks user lifestyle data, predicts stress levels using an machine learning model, and generates personalized schedules based on user preferences. The primary goal of this system is to support work-from-home users in maintaining a healthier and more balanced lifestyle by providing insights and well-structured daily planning

1.3 Background and Motivation

The increasing adoption of remote work has significantly changed the way individuals manage their daily routines. In recent years, a large portion of the global workforce has shifted to work-from-home (WFH) environments, offering flexibility but also introducing challenges related to time management, productivity, and personal well-being. Studies indicate that remote workers often experience difficulties in maintaining structured schedules, balancing work and personal life, and engaging in regular physical activity

Despite the availability of productivity tools such as Google Calendar and Todo list, these systems primarily focus on task organization and scheduling without considering the user's physical condition, emotional state, or lifestyle habits. As a result, users often struggle to maintain consistency in their routines, leading to decreased productivity, poor health habits, and increased stress levels.

Furthermore, modern users face additional challenges such as irregular meal planning, lack of exercise, and difficulty in making everyday decisions efficiently. Research

highlights the growing importance of personalized digital solutions that adapt to individual behavior, preferences, and mental states to improve overall quality of life

The traditional approach to routine management lacks adaptability and personalization, making it ineffective for dynamic work-from-home environments. Motivated by these challenges, this project proposes LifeMate, a smart routine planner designed to integrate productivity, health, and decision-making into a single platform. The system aims to provide personalized scheduling based on user mood and behavior, promote healthier habits, and support users in maintaining a balanced and efficient lifestyle.

1.4 Problem in Brief

Work-from-home (WFH) users often struggle to maintain a structured and balanced daily routine due to the absence of a fixed work environment. This leads to irregular schedules, reduced productivity, poor time management, and neglect of health-related activities such as exercise, proper meals, and rest.

Existing productivity tools mainly focus on task organization and scheduling, but they do not consider the user's physical condition, emotional state, or daily energy levels. As a result, users are unable to adapt their routines based on how they feel, leading to inefficiency and increased stress.

Furthermore, users face difficulty in managing multiple aspects of daily life such as work tasks, meals, fitness, and mental well-being within separate systems. This lack of integration makes routine management complex and less effective. Therefore, there is a need for a unified and intelligent platform that can personalize daily routines and support overall well-being.

1.5 Proposed Solution

For Users:

LifeMate provides an integrated and intelligent platform designed to help work-from-home users manage their daily routines more effectively.

The system offers a smart routine planner that generates personalized schedules based

on user inputs such as mood, energy level, and preferences. This allows users to adapt their daily activities according to their current condition, improving both productivity and well-being. Users can also create and manage tasks manually through a simple and user-friendly interface.

In addition, the platform includes a health and wellness module that supports users by providing hydration reminders, break notifications, and optional wellness tracking features. These features encourage users to maintain healthy habits while working from home.

The system also includes a decision support component that assists users in making everyday choices, such as whether to prepare meals or order food etc. By combining task management, health support, and recommendations, LifeMate helps users achieve a more structured, efficient, and balanced lifestyle.

1.6 Project Aim and Objective

1.6.1 Aim

To develop a routine management system that helps work-from-home users to maintain a balanced lifestyle by analyzing stress using a ML model and generating personalized daily schedules based on user preferences and behavior patterns.

1.6.2 Objectives

1. To develop a user-friendly digital routine management platform

To design and implement a secure, accessible, and interactive platform that enables users to efficiently manage their daily routines through task organization, schedule management, reminders.

2. To implement a personalized scheduling system

To develop a personalized scheduling system that analyze user inputs such as mood, lifestyle patterns, workload, and preferences to generate adaptive schedules, task prioritization, and personalized recommendations using machine-learning based pattern analysis combined with personalized rule-based scheduling logic.

3. To enhance user adaptability through flexible schedule management

To introduce features such as schedule adjustments, rescheduling options, and reminders that allow users to effectively respond to sudden changes such as workload variations, leave requests, or routine disruptions.

4. To promote health and wellness through supportive lifestyle features

To integrate wellness-oriented functionalities including hydration reminders, break notifications, screen-time management, optional health tracking, and meal suggestions to support users' physical and mental well-being while maintaining balanced daily routines.

2. Methodology

The development of the Routine Planner System follows the Agile Software Development methodology using the Scrum framework. Agile Scrum is selected because it supports iterative development, continuous feedback, and flexible requirement handling throughout the project lifecycle. Unlike traditional development approaches, Agile allows the team to enhance the system incrementally while continuously validating requirements with users.

Since the system includes multiple interconnected functionalities such as task and routine management, adaptive scheduling, machine learning-based behavior analysis, personalized recommendation generation, and notification-based reminders, Agile Scrum is particularly suitable. It enables the project to be divided into manageable sprints, ensuring that each feature is developed, tested, and improved step by step.

This approach also helps maintain system quality and usability by incorporating feedback at the end of each sprint, leading to continuous refinement of the scheduling and personalization components.

2.1 Sprint 1: Project Setup & Requirements

Focus:

Foundation of the system and user data collection.

Activities:

- Define system scope, objectives, and user requirements
- Identify user roles (Registered User)
- Design system architecture and database structure
- Set up development environment
- Implement user registration, login, and profile management
- Develop questionnaire module for collecting user inputs (mood, stress, habits, workload)

Outcome:

A working system foundation with user authentication and structured data collection.

2.2 Sprint 2: Stress- Related Pattern Analysis Model

Focus:

Development of AI-based stress analysis.

Activities:

- Define dataset structure provided by the client.
- Preprocess collected questionnaire data
- Train machine learning model for stress prediction
- Classify stress levels (Low, Medium, High)
- Integrate ML model with system backend

Outcome:

A working stress-related pattern analysis model based on user dataset and rule-base classification.

2.3 Sprint 3: Personalized-Rule Based Scheduling System

Focus:

Rule-based routine generation.

Activities:

- Design rule-based scheduling algorithm
- Combine stress level + user preferences (sleep, meals, work hours, breaks)
- Generate personalized daily routines
- Allow manual schedule adjustments
- Integrate scheduling system with frontend interface

Outcome:

A fully functional personalized routine generation system.

2.4 Sprint 4: Wellness Features, Integration & Testing

Focus:

Final system enhancement and deployment readiness.

Activities:

- Implement wellness features (hydration, break, stretch reminders)
- Develop notification system for schedules and alerts
- Perform system integration (Pattern analysis + scheduling + UI)
- Conduct testing and debugging
- Optimize system performance
- Prepare final deployment and documentation

Outcome:

A fully integrated, tested, and deployed intelligent routine management system.

3. Requirements Identification

3.1 Functional Requirements

1. User Registration and Profile Management

- Users can register and log in securely to the system.
- Users can recover accounts using the “Forgot Password” feature.
- Users can create, update, and manage personal profiles.
- Users can set lifestyle preferences such as:
 - Sleep schedule
 - Working hours (fixed/flexible)
 - Meal preferences
 - Workout preferences
 - Free time preferences
 - Reminder preferences
- Users can update profile details and account settings

2. Task and Routine Management

- Users can create, edit, and delete daily tasks and routines.
- Tasks can include:
 - Title
 - Priority level
 - Duration
 - Deadline
 - Category/type
- Tasks can be categorized as:
 - Personal
 - Academic
 - Work
 - Health
 - Others
- Users can mark tasks as completed.
- Users can track daily and weekly productivity progress.
- Users can manually edit generated schedules.
- Users can add sudden leave or unavailable time slots.

3. Personalized Rule-Based Scheduling System (Novelty)

- The system collects user lifestyle and behavioral data.
- The system accepts inputs from users. The system analyzes:
 - Workload
 - Sleep patterns
 - Lifestyle preferences
 - Daily behavior patterns
 - A trained machine learning model predicts user stress levels.
- The system generates personalized routine based on:
 - Identified stress level pattern
 - User preferences
 - User availability
 - user priority
 - Recommendations dynamically adapt according to user changes.

4. Adaptive Scheduling and Rescheduling (Novelty)

- The system generates a schedule using personalized-rule based scheduling.
- Scheduling is based on:
 - User availability
 - Priority rules
 - Stress level
 - Time constraints
- The system allocates:
 - Work sessions
 - Free time
 - Breaks
 - Workout sessions
 - Wellness activities
- Users can:
 - Rearrange schedules
 - Drag and drop activities
 - Accept or reject recommendations
- The system dynamically reschedules activities when:
 - User preferences change

- Sudden leave is added
- Tasks are delayed or removed
- The system inserts intelligent break intervals based on stress level.

5. Wellness and Health Support Module

- The system provides hydration reminders.
- The system provides:
 - Break reminders
 - Stand-up reminders
 - Stretch reminders
 - Screen-time reduction reminders
- Users can enable medicine reminders optionally.
- Users can optionally track:
 - Workouts
 - Water intake
 - Wellness activities
 - Period cycles
- The system promotes work-life balance through wellness suggestions

6. Daily Suggestions

- The system provides suggestions for:
 - Task completion
 - Daily productivity improvement
 - Study/work planning
 - Rest and relaxation planning
 - Meal suggestions
 - Cooking recommendations
 - Food ordering options
 - Suitable workout times
 - Break timing recommendations

- Suggestions aim to:
 - Reduce decision fatigue
 - Improve productivity
 - Maintain healthy daily routines
 - Balance workload and wellness

7. Calendar and Reminder Management

- Users can view schedules using a calendar interface.
- Users can:
 - Add notes
 - Add events
 - Set reminders
 - Manage appointments
- The system sends notifications for:
 - Meetings
 - Deadlines
 - Wellness reminders
 - Scheduled activities
- Users can view:
 - Daily schedules
 - Weekly schedules
 - Upcoming events

3.2 Non-functional Requirements

1. Security and Privacy

The system shall securely store and protect user data and ensure authorized access through secure authentication mechanisms.

2. Performance and Efficiency

The system shall generate schedules, recommendations, reminders, and updates quickly and efficiently with minimal delay.

3. Reliability

The system shall provide stable, accurate, and consistent performance during continuous daily usage and schedule management.

4. Usability

The system shall provide a simple, user-friendly, and interactive interface for managing routines, schedules, reminders, and preferences.

5. Maintainability

The system shall use a modular and scalable architecture to support easy maintenance, updates, and future feature enhancements.

3.3 System Requirements

3.3.1 Hardware Requirements

- Router with an Internet connection
- Laptop / Desktop computer with the following requirements
 - Hard Disk - 10 GB or higher
 - RAM - 4 GB or higher
 - Operating System - Windows 10/11, MacOS
 - Processor – i3 or higher
- Storage Device to store backups, project files, and other related data.

3.3.2 Software Requirements

- **IDE** - Visual Studio Code
- **Package Manager** – npm
- **Database Management Tool** - MongoDB Compass
- **API Testing Tool** – Postman
- **Browser** - Google Chrome
- **Diagram Tools** - draw.io
- **User Interface Design Tool** – Figma

3.3.3 Technologies

- **Front End** – Tailwind CSS, React.js
- **Back End** – Spring Boot (Java)
- **Database** – MongoDB
- **Machine Learning** - Python, Scikit-learn
- **API Communication** - REST API, Flask API
- **Version Control** - Git, GitHub
- **Development Stack** - MERN-inspired architecture with Spring Boot backend (React + Spring Boot + MongoDB)

3.4 User Roles

3.4.1 User (Work-from-Home Professional)

The primary user of the LifeMate system who uses it to manage work tasks, productivity, and daily routine while working remotely.

- Create, update, and manage work tasks and personal routines
- Organize daily/weekly schedules efficiently
- Receive smart reminders and task notifications
- Track productivity and work completion progress
- Follow AI-generated recommendations for better time management and work-life balance
- Adjust schedules based on workload and priorities

3.4.2 Administrator (Admin)

Responsible for managing and maintaining the system for all users.

- Manage user accounts and access permissions
- Monitor system performance and user activity
- Ensure data security and privacy protection
- Maintain and update system features
- Handle technical issues and system support

3.4.3 Visitor (Guest User)

A limited-access user who can explore basic features of the LifeMate system without creating a full account. This role is mainly for first-time users or potential users who want to understand the system before registering.

- View general information about the LifeMate system and its features
- Explore sample dashboards or demo routines (read-only access)
- Understand how AI-based scheduling and productivity tracking works
- Access promotional content or user guides
- Register or upgrade to a full user account (Work-from-Home Professional)
- • Limited or no access to personal data management features

4 Project Plan

4.1 Gantt Chart

Task	Week														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Requirement Gathering & User Research															
System Planning & Architecture Design															
UI/UX Design & Prototype Development															
Database Design & Backend Development															
Authentication & User Profile Module															
Core Task & Routine Management Module															
Machine Learning Model Development															
Scheduling & Adaptive Recommendation System															
Integration, Testing & Debugging															
Deployment & Documentation (Final Phase)															

Figure 1: Gantt Chart

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